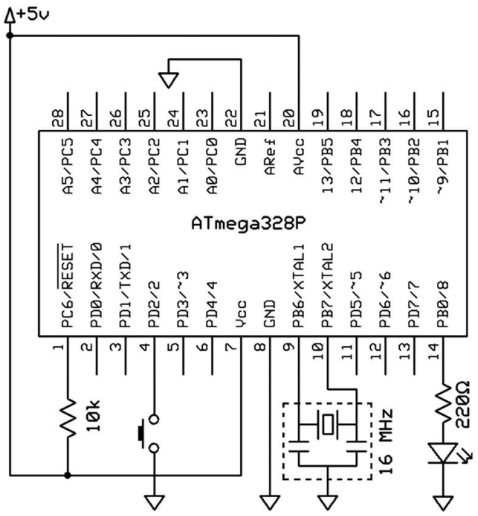
Embedded Systems Development

Exercise 2

1. Consider the following schematic diagram.



Write an assembly language firmware program to blink the LED every 200 ms when the push-button switch is pressed; otherwise, blink it every 400 ms.

2. Consider the following two programs written using the Arduino IDE.

```
// Program 1
#include <avr/io.h>
int main(void)
{
  DDRB = 0x01;
  while (1)
  {
    PORTB = 0x01;
    PORTB = 0x00;
  }
  return 0;
}
```

```
// Program 2
void setup()
{
  pinMode(8, OUTPUT);
}

void loop()
{
  digitalWrite(8, HIGH);
  digitalWrite(8, LOW);
}
```

a. If the two programs are compiled and uploaded to an ATmega328P microcontroller using the Arduino IDE, what would be their intended functionality?

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b. It was observed that, after being compiled and uploaded to an ATmega328P microcontroller using the Arduino IDE, the HEX files of Program 1 and Program 2 consume 142 bytes and 743 bytes of the microcontroller's code ROM, respectively. Explain this observation.

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3. If the following subroutine is assembled and uploaded to an Arduino Uno board, determine the time delay it generates in milliseconds. Show all your workings clearly.

```
DELAY: LDI R16, 0x69
LOOP1: LDI R17, 0xFD
LOOP2: DEC R17
       BRNE LOOP2
       DEC R16
       NOP
       BRNE LOOP1
       RET
```

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